Debaditya Hait

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Software Development Engineer | Backend, Full-Stack & Distributed Systems

Summary

Computer Science undergraduate building end-to-end web products, API-backed systems and cloud-native services with attention to correctness, persistence and failure recovery. Implemented multi-tenant monitoring, local-first editing and marketplace comparison; contributed focused fixes to Redis, Vite, pnpm and distributed-query open source.

Education

SRM Institute of Science and Technology 2023-Present
B.Tech., Computer Science and Engineering – CGPA: 8.5/10 *Kattankulathur, India*

Indian Institute of Technology Madras 2024-Present
B.S., Data Science and Applications – Online degree program *Chennai, India*

Technical Skills

Languages & CS: Python, TypeScript/JavaScript, C/C++, Java, SQL, data structures and algorithms

Backend & data: Node.js, FastAPI, Hono, REST APIs, PostgreSQL, Prisma, SQLAlchemy, Redis/BullMQ, SQLite

Cloud & systems: Cloudflare Workers/D1/Queues, Azure, Docker, gRPC/protobuf integration, event-driven jobs, idempotency, CI/CD

Frontend & state: React, Next.js, Vite, Konva, IndexedDB/OPFS, offline synchronization

Testing & security: Vitest, Playwright, GitHub Actions, tenant isolation, API authorization, AES-GCM

Experience

Samsung R&D Institute India Apr 2025-Sep 2025
PRISM Research Intern — LLM Security *Remote*

- Fine-tuned a 395M-parameter ModernBERT classifier on 75,000+ augmented benign and adversarial prompts for prompt-injection and jailbreak detection in LLM and agent workflows.
- Achieved 92.70% accuracy and 0.9021 F1 on a 315-prompt adversarial benchmark; measured 35 ms encoder inference, about 300x faster than benchmarked generation-based alternatives.

Indian Institute of Technology Madras Jun 2025-Aug 2025
AI Research & Prototyping Intern *On-site, Chennai*

- Developed physics-informed neural networks in PyTorch to model Alzheimer's progression from longitudinal MRI features and biological differential-equation constraints.
- Applied automatic differentiation, collocation sampling and adaptive loss weighting; reported 27% higher early-prediction accuracy versus research deep-learning baselines.

Selected Projects

Pulseflare | *Multi-tenant monitoring and incident response* | GitHub | Website
TypeScript / Hono / Cloudflare Workers/D1/Queues / React

- Built an HTTP uptime and job-heartbeat monitoring platform with incident timelines, Slack/Discord/Telegram/webhook alerts, public status feeds, deployment annotations and private evidence-grounded AI incident chat.
- Coordinated scheduled checks and notifications with atomic D1 leases, a transactional outbox, separate queues, tenant-scoped access and retry/idempotency controls; 106 unit/integration tests passed.
- Designed scoped API keys, encrypted webhook credentials and private tenant-bound incident evidence; status/incident views are backed by bounded retention, quotas and cached public feeds.

ShopLense | *Location-aware product comparison and alerts* | GitHub | Website
Next.js / TypeScript / Node.js / PostgreSQL/Prisma

- Developed self-hosted price/stock comparison for Zepto, Blinkit, Flipkart (including Flipkart Minutes) and Swiggy Instamart, with location-specific offers, approximate product matching and saved price/restock rules.
- Built concurrent web-session marketplace adapters, failure-isolated API queries and a polling worker for Web Push/ntfy notifications; 44 application regression tests passed.

Kanvas | *Local-first graphic and photo editor* | GitHub | Website
React / TypeScript / Konva / IndexedDB/OPFS / Supabase

- Built a browser design editor for layered objects, nondestructive photo adjustments, multipage compositions, offline projects and PNG/JPEG/WebP/PDF exports.
- Implemented durable local persistence, revision-aware cloud sync, conflict copies and revocable sharing; owner-reported 6,000+ monthly users and 25,000+ designs created.
- Used a validated document schema, shared rendering logic and cross-browser validation to maintain editor/export consistency; 46 unit/integration tests passed.

OnyxBin | *Self-hosted Telegram-backed file workspace* | GitHub | Website
Next.js / TypeScript / NestJS/Go integration / PostgreSQL / Redis

- Developed a file-management workspace with nested folders, previews, chunked upload/download queues, revocable shares, trash recovery, ZIP operations and S3-compatible credential flows.
- Integrated the workspace with existing NestJS/Go storage services, PostgreSQL metadata and Telegram chunk storage; validated frontend/API wiring in 18 fixture-backed browser scenarios.
- Exposed supported S3-compatible object and multipart workflows through existing backend services, with signature-based credentials and presigned requests; no full-S3 compatibility certification claimed.

Open-Source Engineering

Contributed upstream engineering fixes to infrastructure and developer-tooling repositories, including Node Redis, pnpm, Vite, Apache DataFusion Ballista and PostHog. Statuses below reflect the October 2026 contribution report.

Node Redis (PR #3388) | *Merged (Oct 2026 source record)* | *TypeScript / Redis*

- Bounded Sentinel topology-rediscovery retries and backoff after reconnect, preventing unbounded retry loops and socket exhaustion on partitions.

pnpm (PR #13059) | *Merged (Oct 2026 source record)* | *Rust*

- Fixed Git dependency resolution in the Rust-native package-manager engine by reading package names from repository manifests.

Vite (PR #22947) | *Merged (Oct 2026 source record)* | *TypeScript / build tools*

- Restored CSS chunk entries in dynamic-import maps so production builds can preload associated styles; added regression coverage.

Apache DataFusion Ballista (PR #2397) | *Merged (Oct 2026 source record)* | *Rust / OpenAPI*

- Exposed and tested machine-readable OpenAPI documentation for distributed query scheduler routes and DTO schemas.

PostHog (PR #70314) | *Merged (Oct 2026 source record)* | *Rust / Rayon*

- Made sourcemap-upload concurrency configurable via an isolated thread pool and option validation.

Competitive Programming

- Codeforces: Expert (current); previously held Candidate Master at peak before returning to Expert.
- LeetCode: Guardian; top 1%.
- CodeChef: 5-star.

Publications, Achievements & Certifications

- **Achievement:** Goldman Sachs India Hackathon 2026 (Computer Science): all-India rank 144 of 15,953.
- **Achievement:** Flipkart GRiD 8.0 (2026), Software Development: national semifinalist.
- **Achievement:** Goldman Sachs India Hackathon 2026 (Quant): all-India rank 42 of approximately 14,000.
- **Certification:** Microsoft Certified: DevOps Engineer Expert (earned Oct 2026).
- **Certification:** Microsoft Certified: Azure Developer Associate (earned Jun 2025).
- **Certification:** GitHub Actions (GitHub-issued; earned Oct 2026).
- **Certification:** GitHub Advanced Security (GitHub-issued; earned Oct 2026).